
3. Prepare Test Plan

3.1 Introduction

The Test Plan is a formal document that defines the overall strategy, approach, scope, resources, schedule, and deliverables for the testing activities of a software application.

In this project, the Test Plan is prepared for the **OWASP Juice Shop Web Application**, which is a deliberately insecure application used for security testing and training purposes. The primary focus of this test plan is to perform structured **security testing and automation testing** to identify vulnerabilities such as SQL Injection, Cross-Site Scripting (XSS), authentication issues, and input validation flaws.

The test plan also defines how the **automation framework built using Selenium WebDriver and TestNG** will be used to execute security test cases in a systematic and repeatable manner.

3.2 Objectives of Test Plan

The main objectives of this Test Plan are:

- To define a structured approach for security testing of OWASP Juice Shop
 - To identify security vulnerabilities in the application
 - To validate application behavior against malicious inputs
 - To execute automated test cases using Selenium WebDriver and TestNG
 - To ensure maximum test coverage of security scenarios
 - To define testing scope, strategy, and environment setup
 - To ensure proper reporting of defects and test results
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3.3 Scope of Testing

In Scope

The following areas are included in the testing scope:

- Authentication and Authorization testing
- SQL Injection testing
- Cross-Site Scripting (XSS) testing
- Input validation testing

- Session management testing
 - Error handling validation
 - Security test automation using Selenium WebDriver
 - Execution of TestNG test suites
 - Generation of test reports and screenshots
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Out of Scope

The following areas are excluded from testing:

- Source code modification of the application
 - Server-side infrastructure security testing
 - Network penetration testing beyond application layer
 - Performance and load testing (unless security-related)
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3.4 Test Strategy

The testing strategy used in this project is a combination of **manual security testing and automation testing**.

1. Manual Testing Strategy

- Identify security vulnerabilities by manually injecting payloads
 - Observe application response to malicious inputs
 - Validate error messages and system behavior
 - Record findings for documentation
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2. Automation Testing Strategy

- Use Selenium WebDriver for browser automation
 - Use TestNG for test execution and grouping
 - Implement Page Object Model (POM) for maintainability
 - Execute repeated security test cases automatically
 - Validate security behavior using assertions
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3. Security Testing Approach

The following security testing techniques are applied:

- Input-based attack testing (SQL Injection, XSS)
- Negative testing with invalid inputs

- Authorization bypass testing
 - Session validation testing
 - Error message validation testing
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3.5 Tools and Technologies Used

- Java (Programming Language)
 - Selenium WebDriver (Automation Tool)
 - TestNG (Testing Framework)
 - Maven (Build Tool)
 - Eclipse IDE
 - Chrome Browser
 - OWASP Juice Shop Application
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3.6 Automation Framework Overview

The automation framework is designed using **Page Object Model (POM)** architecture.

Framework Layers:

Base Layer

- BaseTest → Handles browser setup and teardown
- DriverFactory → Manages WebDriver instances

Page Layer

- LoginPage
- HomePage
- FeedbackPage

Test Layer

- Authentication_Test
- SQLInjection_Test
- XSS_Test

Utility Layer

- ConfigReader
- ScreenshotUtil
- ExcelReader

Listener Layer

- TestListener → Handles reporting and logging

Benefits of Framework:

- Easy maintenance
 - Code reusability
 - Scalability
 - Better debugging support
 - Structured test execution
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3.7 Test Environment

- Operating System: Windows 10
 - Browser: Google Chrome
 - Automation Tool: Selenium WebDriver
 - Framework: TestNG
 - IDE: Eclipse
 - Application: OWASP Juice Shop (Local/Hosted)
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3.8 Entry Criteria

Testing will start only when the following conditions are met:

- Application is deployed and accessible
 - Test environment is stable
 - Test cases and scenarios are prepared
 - Automation framework is ready
 - Required dependencies are installed (Java, Maven, Selenium)
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3.9 Exit Criteria

Testing will be considered complete when:

- All planned test cases are executed
 - No critical security defects remain open
 - Test reports are generated successfully
 - Automation execution is stable
 - Screenshots and evidence are collected
 - Test closure report is prepared
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3.10 Risks and Assumptions

Risks:

- Application may behave differently under repeated attack payloads
- Browser compatibility issues during automation
- Dynamic UI elements may cause locator failures
- Test failures due to unstable environment

Assumptions:

- Application is accessible during testing
 - Test environment remains stable
 - Security vulnerabilities exist as expected (since Juice Shop is intentionally vulnerable)
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3.11 Deliverables

- Test Plan Document
 - Test Scenarios Document
 - Test Cases Document
 - Automation Framework Code
 - Test Execution Reports
 - Bug Reports (if any)
 - TestNG HTML Reports
 - Screenshots of execution
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3.12 Conclusion

The Test Plan provides a clear roadmap for executing security testing of the OWASP Juice Shop application. It ensures that testing activities are structured, repeatable, and aligned with security testing objectives.

The combination of manual and automation testing improves test coverage and ensures efficient identification of vulnerabilities.
